Cross spectra between thin redshift bins

Which pairs the data vector measures, and how the
Fingers-of-God blur changes their overlap

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1 Summary

The questions. The forecast cuts the DESI galaxies into thin redshift bins and measures the angular power spectrum of every bin. For some pairs of bins it also measures their cross spectrum, the correlation between the two bins' maps. Four bin layouts slice the galaxies: $\Delta d_{\rm com} = 62$ Mpc, 76 bins of equal comoving depth, and three of fixed redshift width, $\Delta z = 0.02$, 0.05 and 0.2, with 90, 36 and 10 bins. The first two are the thin layouts. This note answers three questions:

1. How does the Fingers-of-God (FoG) blur, at its default strength, change how much neighbouring bins overlap?
2. Which pairs of bins have their cross spectrum in the data vector, and why those?
3. What does that rule mean in redshift, on each of the four bin layouts?

The answers.

1. The blur has a width of 5–6 Mpc along the line of sight, about a tenth of a thin bin. It raises the overlap of two neighbouring thin bins from about 0.003, the one grid step where their tabulated edges meet, to about 0.06. Bins one apart stay disjoint. On $\Delta d_{\rm com} = 62$ Mpc the blur makes ten LRG3–ELG1 pairs overlap by 0.03, half as much as a neighbour pair, although their raw ranges just miss; the data vector measures them too (Figure 1).
2. A pair is measured when the two bins' redshift distributions $n(z)$ overlap or touch, either as measured or once blurred by the FoG effect of item 1. That gives the neighbours along each galaxy sample's chain of bins, plus the pairs of an LRG bin and an ELG bin at about the same redshift: at $0.8 < z < 1.1$ the LRG3 and ELG1 groups are two galaxy samples observing one volume (Figures 2 and 3). The four layouts measure 107, 133, 52 and 13 cross spectra.
3. On the layouts of fixed redshift width only adjacent bins pair. On $\Delta d_{\rm com} = 62$ Mpc the rule is adjacency in comoving distance, which no single cut in redshift can express: a bin of fixed comoving depth is 0.015 wide in redshift at $z = 0.1$ and 0.035 at $z = 1.5$.

2 Fiducial model and parameter values

Three of this note's figures (Figures 1, 2 and 3) depend only on the shipped redshift distributions, the pair lists and the default velocity dispersions: no scale cut and no bias value enters

them. Only the pair-threshold test of Section 10 runs a Fisher fit. This section freezes the values behind both, so the record does not depend on configuration files that can change.

Geometry of the pair rule and the masks. The comoving distances of the $\Delta d_{\rm com} = 62$ Mpc slicing, of the pair rule and of the shipped masks use $\chi(z)$ of a flat ΛCDM with $H_0 = 67.32$ km/s/Mpc and $\Omega_m = 0.3$; the masks stop every bin at $\ell + 1/2 \le k\,\chi(\bar{z}_i)$ with $k = 0.1185\,\mathrm{Mpc}^{-1}$ (about $0.176\,h$/Mpc), with no growth scaling.

Layouts and pairs. The four layouts hold 76, 90, 36 and 10 lens bins and measure 107, 133, 52 and 13 cross pairs ($\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2; Tables 1 and 3). The paper data sets carry these pair lists; the exploration examples keep the older 97-pair `*_linear` files on $\Delta d_{\rm com} = 62$ Mpc (Section 6).

Velocity dispersions. The fiducial dispersions are 371, 358, 357, 348, 343, 304 and 289 km/s for the seven groups (BGS split at $\bar{z} = 0.255$), with exploration prior widths 145, 105, 70, 50, 50, 40 and 30 km/s and paper widths six times these (Table 2 and Section 5). Every blur in this note runs at the fiducials; only the fit below uses the widths, the exploration ones.

The pair-threshold fit. Its cosmology is the evaluate override of the example yamls: $10^9 A_s = 2.1$ (A_s the amplitude of the primordial curvature power spectrum), $n_s = 0.96605$, $H_0 = 67.32$ km/s/Mpc, $\Omega_b = 0.04$, $\Omega_m = 0.3$, $w_0 = -1$, $w_a = 0$, $\sum m_\nu = 0.06$ eV and $\tau = 0.0697186$ fixed; the derived $\sigma_8 = 0.8278$. The linear bias is $b_{1,i} = B_g / D(\bar{z}_i)$ with $B_g = 1.34$ (BGS), 1.70 (LRG1–LRG3) and 0.84 (ELG1, ELG2), under the pessimistic prior (3% correlated residuals, $L = 0.05$, a free amplitude and tilt per group; `notes/correlated_bias_prior_v2`). The higher-order terms are model B + tilts (`notes/higher_order_bias`): offsets Δ_2 and Δ_s of widths 2 and 1, counterterms α_0 and α_2 of width $0.5/k^2_{\rm max}$ at each cut, stochastic amplitudes SN_0 and SN_2 of widths 2 and 5, and flat tilts, with α_2 free and the exploration σ_v widths: the reference model of `notes/fog_parameter_degeneracy_study`, not the paper model, which fixes $\alpha_2 = 0$ in every bin and widens every σ_v prior sixfold.

Scale cut and covariance of the fit. The fit keeps the bands of the UV-fraction cut computed on the paper model's fiducial (`chains/EXAMPLE_EVALUATE_FISHER_MODEL_SCALE_CUTS.pkl`; a band is kept while less than 5% of it comes from the power spectrum above $k_{\rm max}/D(\bar{z}_i)$), intersected with the shipped mask. The covariance is the Gaussian covariance of the 107-pair data set, built on the one-loop fiducial (b_2 and b_{s^2} on their calibrations) with massless neutrinos over 14,000 deg^2, its inverse rescaled by 12,300/14,000.

3 Why cross spectra between thin bins matter

The angular power spectrum. Take the galaxies of bin i and draw a map of their number on the sky: $\delta_i(\hat{n})$ is the fractional excess of galaxies in the direction $\hat{n}$ over the mean. Expand the map in spherical harmonics, $\delta_i(\hat{n}) = \sum_{\ell m} a^i_{\ell m} Y_{\ell m}(\hat{n})$. The multipole ℓ labels an angular scale of about $180°/\ell$: $\ell = 20$ is about 9°, $\ell = 500$ about 0.4°.

The auto spectrum of bin i is the variance of its coefficients, $C^{ii}_\ell = \langle |a^i_{\ell m}|^2 \rangle$; the cross spectrum of bins i and j is their covariance, $C^{ij}_\ell = \langle a^i_{\ell m} a^{j*}_{\ell m} \rangle$. Two unrelated maps have a cross

spectrum that averages to zero; two maps of the same structures have a positive one. The data vector holds these spectra in 53 bands of width $\Delta\ell = 10$ over $15 \leq \ell \leq 544$.

Why neighbouring thin bins correlate. A thin bin is a shell of the universe a few tens of Mpc deep: 62 Mpc on $\Delta d_{\rm com} = 62$ Mpc, and between 37 and 85 Mpc for $\Delta z = 0.02$. The matter distribution has structures of every size. One that is 200 Mpc across spans three or four such shells, and each of them sees it in the same direction on the sky. Their maps share it, so the cross spectrum of two neighbouring shells is large at the angular scale of that structure. A structure a few Mpc across sits in one shell only, unless it sits where two shells meet.

Limber and non-Limber. These two cases are the two regimes of a cross spectrum. At high ℓ the Limber approximation [1] holds: only Fourier modes perpendicular to the line of sight contribute, and

$$C_\ell^{ij} \;\simeq\; \int d\chi \, \frac{W_i(\chi)\, W_j(\chi)}{\chi^2} \, P\!\left(k = \frac{\ell + 1/2}{\chi}\right), \tag{1}$$

with χ the comoving distance, $W_i(\chi)$ the redshift distribution of bin i written per unit comoving distance, and $P(k)$ the power spectrum of the galaxy density at wavenumber k. Two shells whose W do not overlap then have no cross spectrum. At low ℓ the approximation fails for thin shells: modes with a component along the line of sight longer than the shell depth connect shells that do not touch. The forecast computes every spectrum without the Limber approximation, so it keeps that low-ℓ correlation.

Where the radial information sits. Slicing a sample into thin shells keeps information along the line of sight: a shell of depth $\Delta\chi$ resolves line-of-sight wavenumbers up to about $\pi/\Delta\chi$, $0.05\,\mathrm{Mpc}^{-1}$ for 62 Mpc. The auto spectra say how clustered each shell is; the cross spectra say how the clustering of one shell continues into the next.

How much the cross spectra are worth depends on the bin layout: the thinner the shells, the more of the line-of-sight information sits in the cross spectra between neighbours rather than in the auto spectra.

4 The six groups and the LRG–ELG overlap

Three galaxy samples. DESI selects three kinds of galaxies, each by its own colours and brightness, and each fills the redshifts where it is bright and numerous enough to observe:

- **BGS**, the Bright Galaxy Survey: the brightest galaxies at low redshift, here $0.1 < z < 0.4$;
- **LRG**, luminous red galaxies: massive galaxies that stopped forming stars, $0.4 < z < 1.1$;
- **ELG**, emission-line galaxies: star-forming galaxies, found by the bright emission lines of their gas, $0.8 < z < 1.6$.

Nothing forces these ranges to tile the redshift axis. LRG and ELG both cover $0.8 < z < 1.1$.

Six groups. The forecast splits the samples into six groups by the bins' mean redshift $\bar{z}_i$, and gives each group its own bias parameters (Table 1): BGS whole; LRG1 ($\bar{z}_i < 0.6$), LRG2 ($0.6 \le \bar{z}_i < 0.8$) and LRG3 ($\bar{z}_i \ge 0.8$); ELG1 ($\bar{z}_i < 1.1$) and ELG2 ($\bar{z}_i \ge 1.1$). LRG3 and ELG1 are therefore two different galaxy populations in the same volume, at $z \approx 0.8$–1.1.

group	sample	bins	$\bar{z}_i$
BGS	bright galaxies	1–18	0.11–0.38
LRG1	luminous red galaxies	19–29	0.41–0.60
LRG2	luminous red galaxies	30–38	0.62–0.79
LRG3	luminous red galaxies	39–49	0.81–1.06
ELG1	emission-line galaxies	50–61	0.81–1.09
ELG2	emission-line galaxies	62–76	1.11–1.55

Table 1: The six groups of $\Delta d_{\rm com} = 62$ Mpc, with bins numbered from 1 in data-vector order. LRG3 and ELG1 cover the same redshifts.

The numbering runs group by group. Bins are numbered in the order of the data vector: all BGS bins by increasing redshift, then all LRG bins, then all ELG bins. The numbering therefore jumps back in redshift between bins 49 and 50: bin 49 (LRG3) has $\bar{z} = 1.060$, bin 50 (ELG1) has $\bar{z} = 0.812$. Two bins far apart in number can sit at the same redshift: LRG3 bin 44 has $\bar{z} = 0.932$ and ELG1 bin 55 has $\bar{z} = 0.931$.

What the LRG–ELG cross spectrum measures. An LRG bin and an ELG bin at the same redshift sample the same matter with different galaxies. On large scales a galaxy map is the matter map times a number, the linear bias b_i of the bin, so their cross spectrum is about $b_i b_j$ times the matter spectrum, while the two autos are about b_i^2 and b_j^2 times it. The three spectra together tie the two biases to each other. Two tracers of the same structures also share the random fluctuations of those structures, so comparing them measures their bias ratio with less of that randomness than either map alone (the multi-tracer idea [2]).

Shot noise and the stochastic terms. A galaxy map is made of discrete objects, so even a random scatter of galaxies has a nonzero auto spectrum: the shot noise, $1/\bar{n}_i$ for a bin of $\bar{n}_i$ galaxies per steradian. A cross spectrum has no shot noise, because no galaxy is in two bins: within one sample each galaxy has one measured redshift, and two samples are different galaxies.

The one-loop model adds stochastic terms beyond this Poisson noise, SN_0 and SN_2. They are properties of the three-dimensional galaxy field, so they reach the cross spectrum of two bins of one sample through the overlap of their kernels, $\int d\chi\, W_i W_j/\chi^2$ for SN_0 and the same integral of the kernels' radial derivatives for SN_2 (`stochastic_templates` in `likelihood/desi_cl.py`). An LRG–ELG pair gets none: two different populations do not share a stochastic term, and the model has no separate cross-stochasticity parameter.

5 The Fingers-of-God blur

5.1 What it is

Redshift is not distance. The forecast places each galaxy by its measured redshift. That redshift has two parts: the cosmological redshift, set by the galaxy's distance, and a Doppler shift from the galaxy's own velocity v along the line of sight, its peculiar velocity. The two multiply, $1 + z_{\rm obs} = (1 + z)(1 + v/c)$, so for $v \ll c$

$$z_{\rm obs} - z \;\simeq\; (1 + z)\,\frac{v}{c}. \tag{2}$$

For example, $v = 350$ km/s at $z = 0.5$ shifts the redshift by 1.75×10^{-3}. In distance this is $\delta\chi = c\,\delta z/H(z) = (1 + z)\,v/H(z) \approx 6$ Mpc, with $H(0.5) = 88$ km/s/Mpc in the examples' cosmology ($H_0 = 67.32$ km/s/Mpc, $\Omega_m = 0.3$).

Fingers of God. Inside groups and clusters, galaxies orbit their common halo with random speeds of several hundred km/s. In a map that places every galaxy at its redshift distance, a cluster is stretched along the line of sight into a long finger pointing at the observer: the Fingers of God [3]. The orderly infall of galaxies onto large structures is a different effect, the Kaiser distortion [4], which the model computes separately.

What it does to a thin bin. A bin collects the galaxies whose measured redshift falls inside its range: a sharp cut in $z_{\rm obs}$. The matter that the galaxies trace sits at their true distances. A galaxy just outside the range, moving towards it, is measured inside; one just inside, moving away, is measured outside. So the galaxies of the bin trace the matter over a slightly wider, softer-edged range than the histogram of their measured redshifts. The same convolution describes a sharp shell in true redshift observed blurred along the line of sight; for a symmetric velocity distribution the two descriptions agree.

5.2 The model in the likelihood

The likelihood replaces each bin's raw $n_i(z)$ by its convolution with a Gaussian in redshift [5],

$$n_i^{\rm FoG}(z) = \int dz'\, n_i(z')\,\frac{1}{\sqrt{2\pi}\,\sigma_{z,i}}\,\exp\!\left[-\frac{(z - z')^2}{2\sigma_{z,i}^2}\right], \qquad \sigma_{z,i} = (1 + \bar{z}_i)\,\frac{\sigma_v}{c}, \tag{3}$$

with σ_v the velocity dispersion of the bin's group, the standard deviation of the line-of-sight velocities, and $c = 299792.458$ km/s. The width follows from Eq. (2).

The code is `set_lens_related` in `likelihood/desi_cl.py`. It convolves a copy of each raw histogram with `scipy.ndimage.gaussian_filter1d` on the $n(z)$ table's grid, the DESI grid refined five-fold (step 7.0×10^{-5}, so $\sigma_{z,i}$ spans 20 to 36 steps), with the width in grid steps and zero beyond the grid ends. The Gaussian is truncated at $4\sigma_{z,i}$, scipy's default. The σ_v are sampled parameters, so the convolution runs again at every sampled point. The figures of this note repeat that call; at the default σ_v the result equals, bit for bit, the convolved tables that the covariance adapter writes with the same call (`covariance/<layout>_fog.nz`).

5.3 Where the velocity dispersions come from

The source forecast measured the line-of-sight velocity dispersion of its simulated galaxies as a function of redshift, with an error bar per bin. The project reads one value per group off that measurement (Table 2). The measured curve steps inside BGS, between the bins at $\bar{z} = 0.247$ and 0.263, so BGS takes two values, split at $z = 0.255$: seven dispersion groups for six bias groups.

group	σ_v [km/s]	exploration width	paper width	$\sigma_{z,i}$ [10^{-3}]	$\delta\chi$ [Mpc]
BGS, $\bar{z}_i < 0.255$	371	145	870	1.37–1.54	5.8–6.1
BGS, $\bar{z}_i \geq 0.255$	358	105	630	1.51–1.65	5.9–6.0
LRG1	357	70	420	1.68–1.90	6.0–6.1
LRG2	348	50	300	1.88–2.08	6.0
LRG3	343	50	300	2.07–2.36	5.8–5.9
ELG1	304	40	240	1.84–2.12	5.1–5.2
ELG2	289	30	180	2.04–2.46	4.6–4.8

Table 2: The default velocity dispersion of each group, the widths of its Gaussian prior in the exploration files and in the paper model (six times the exploration width; both in km/s), and the resulting redshift width $\sigma_{z,i}$ and comoving width $\delta\chi = (1 + \bar{z}_i)\,\sigma_v/H(\bar{z}_i)$ over the bins of $\Delta d_{\rm com} = 62$ Mpc.

Each prior is a Gaussian centred on the default and truncated at zero. The exploration width is five times the per-bin error bar of the measurement; the paper model multiplies every width by six, so that the sampled dispersions can absorb the radial counterterm it drops (`notes/fog_parameter_degeneracy_study`). At the sixfold width the truncation matters for the first BGS group: it removes 33% of the Gaussian, and the truncated mean is about 850 km/s, while the Fisher forecasts treat every σ_v prior as an untruncated Gaussian about the default. The exploration defaults and widths live in `likelihood/params_desi*.yaml` (the `DESI_SIGV` parameters; the first bin of each group carries the prior and the others are tied to it), the paper widths in `likelihood/params_desi*_bias_pessimistic.yaml`, and the same seven defaults in `CHUNK_SIGMA_V` of `covariance/lssty1xdesi_covariance.py`, which convolves the covariance's kernels at fixed σ_v. Every figure of this note blurs at the defaults; the fit of Section 10 uses the exploration widths (Section 2). The ELG dispersions are the lowest, as expected for star-forming galaxies, which tend to live in lighter halos with slower orbits.

5.4 How large the blur is

A nearly constant width in distance. In comoving distance the blur is $\delta\chi = (1 + z)\,\sigma_v/H(z)$ (last column of Table 2): 5.8–6.1 Mpc for BGS and LRG and 4.6–5.2 Mpc for ELG, at every redshift. It barely changes with z because $(1 + z)/H(z)$ changes by at most 10% over $0.1 < z < 1.6$.

Against the bin width. What matters for overlap is $\sigma_{z,i}$ divided by the bin's width w_i, the redshift range its galaxies were selected in. On $\Delta d_{\rm com} = 62$ Mpc it is the same ratio in distance, $\delta\chi/62$ Mpc: 0.069–0.096, about a tenth of the bin at every redshift. On $\Delta z = 0.02$ it grows with redshift, from 0.069 to 0.125, because a fixed Δz is a thinner shell at high z. On $\Delta z = 0.05$ it is 0.028–0.050, and on $\Delta z = 0.2$ 0.007–0.023 (Figure 1, top right).

A Gaussian blur of a sharp edge moves a fraction of about $\sigma_z/(\sqrt{2\pi}\,w)$ of a bin's galaxies across each edge. The blurred $n(z)$ of a thin bin has 5–8% of its weight outside its raw range on $\Delta d_{\rm com} = 62$ Mpc and 5–10% on $\Delta z = 0.02$, against 2–4% on $\Delta z = 0.05$ and 0.5–2% on $\Delta z = 0.2$. That weight lies inside the neighbouring bins.

5.5 What it does to the overlap

A measure of overlap. For two bins define the normalized overlap integral

$$O_{ij} \;=\; \frac{\int dz\, n_i(z)\, n_j(z)}{\sqrt{\int dz\, n_i^2(z)\, \int dz\, n_j^2(z)}}, \tag{4}$$

1 for two identical distributions and 0 for disjoint ones, whatever their normalization. It has a direct meaning. In the Limber regime, Eq. (1), the power spectrum and χ barely change across two thin bins, so O_{ij} is close to the correlation coefficient $C_\ell^{ij}/\sqrt{C_\ell^{ii}C_\ell^{jj}}$ of the two bins at high ℓ. The SN_0 cross term of Section 4 is nearly the same integral.

Neighbours. Without the blur, two neighbouring bins touch only over the one grid step where their tabulated edges meet, and their overlap is about $h/(6\,w)$: 0.0015–0.0039 on $\Delta d_{\rm com} = 62$ Mpc, 0.0027–0.0029 on $\Delta z = 0.02$, 0.0011–0.0012 on $\Delta z = 0.05$ and 0.0003–0.0004 on $\Delta z = 0.2$. This number measures the grid, not the galaxies. With the blur at the default σ_v the edges become ramps about $4\sigma_z$ wide, and the overlap becomes about $0.6\,\sigma_z/w$: 0.039–0.061 on $\Delta d_{\rm com} = 62$ Mpc (median 0.057), 0.043–0.083 on $\Delta z = 0.02$ (0.066), 0.018–0.032 on $\Delta z = 0.05$ (0.024) and 0.006–0.009 on $\Delta z = 0.2$ (0.007). On the thin layouts this is a 6% correlation at high ℓ that the raw $n(z)$ would put at 0.3%.

Bins one apart stay disjoint. The bin between them is at least $8\sigma_z$ wide, and the Gaussian is truncated at $4\sigma_z$, so the blurred $n(z)$ of two bins one apart do not meet. The only exception is the last ELG2 bins of $\Delta z = 0.02$, where $\sigma_z/w = 0.125$ and the truncated tails touch, with an overlap below 2×10^{-14}. Even an untruncated Gaussian gives two blurred top hats one bin apart a percent-level overlap only at $\sigma_z \approx 0.4\,w$, four times the default on the thin layouts. Within two prior widths of every default σ_z/w stays below 0.2, where that overlap is below 2×10^{-5}.

6 Which pairs are measured

The first step: raw ranges. Each bin's raw $n(z)$ is the histogram of the measured redshifts of its galaxies, tabulated on the DESI redshift grid of step $h = 3.5\times10^{-4}$. Let $z_i^{\rm lo}$ and $z_i^{\rm hi}$ be the first and last grid redshifts where $n_i(z)$ is nonzero. Bins i and j pair when their ranges overlap or touch:

$$z_j^{\rm lo} \le z_i^{\rm hi} + 1.5\,h \quad\text{and}\quad z_i^{\rm lo} \le z_j^{\rm hi} + 1.5\,h. \tag{5}$$

The slack of 1.5 grid steps is there because two neighbouring bins of one sample either share their edge node or sit one node apart; 1.5 steps catches both and nothing farther.

For example, on $\Delta z = 0.02$ bin 1 ends at $z = 0.1198$ and bin 2 starts one grid step later, at 0.1202: they pair. Bin 3 starts at 0.1401, 0.020 after the end of bin 1: bins 1 and 3 do not pair.

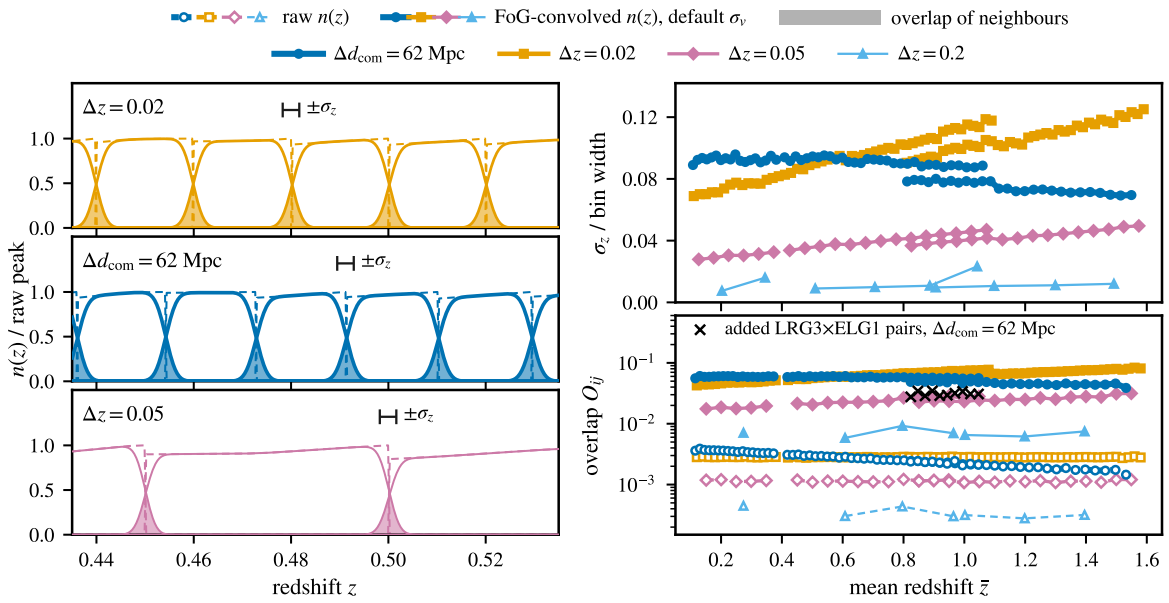


Figure 1: The Fingers-of-God blur at the default σ_v. Left: raw (dashed) and blurred (solid) $n(z)$ of the LRG1 bins in $0.435 < z < 0.535$ on three layouts, each divided by its raw peak; shaded: the overlap $\min(n_i, n_j)$ of neighbouring blurred bins; bar: $\pm\sigma_z$. Right: σ_z over the bin width, and the overlap O_{ij} of Eq. (4) for consecutive bins of one sample, raw (dashed, open markers) and blurred (solid, filled), against the mean redshift. Crosses: the ten LRG3–ELG1 pairs of $\Delta d_{\rm com} = 62$ Mpc that the second step of the pair rule (Section 6) adds, whose raw overlap is zero. Bins one apart overlap by at most 2×10^{-14}, raw or blurred.

The second step: blurred kernels. The model never uses the raw $n(z)$ directly. It first blurs each one by the Fingers-of-God effect of Section 5, which spreads every bin a few Mpc beyond its edges. Two bins whose raw ranges just miss can therefore still overlap once blurred. The second step adds every such pair: bins i and j also pair when their blurred $n(z)$ overlap at all, $O_{ij} \geq 10^{-6}$ at the default velocity dispersions, with O_{ij} the normalized overlap integral of Eq. (4) (`scripts/make_desi_pairs.py`).

The threshold is not a tuning. Before this step the blurred overlaps of unmeasured pairs fall into two sets far apart: ten pairs at 0.028–0.035, and every other pair below 2×10^{-14}, the reach of the truncated Gaussian tails. Any threshold between 10^{-6} and 0.0277 gives the same lists: those ten pairs on $\Delta d_{\rm com} = 62$ Mpc, none on the fixed-Δz layouts.

Where the pair lists come from. For the three fixed-Δz layouts the first step writes the pair list (`cross_pairs` in `scripts/make_desi_dz_case.py`), and the second adds nothing. On $\Delta d_{\rm com} = 62$ Mpc the first list comes from the DESI forecast's data file (a `sacc` file), which pairs bins by redshift proximity; `scripts/make_desi_dataset.py` imports it. The first step reproduces 96 of its 97 pairs. The exception is the boundary pair of BGS bin 18 and LRG1 bin 19: BGS ends at $z = 0.390$ and LRG starts at $z = 0.400$, so their ranges are 0.010 apart, and the forecast includes them anyway; the list keeps it. Blurred, their kernels overlap by only 3×10^{-7}: the ranges are six times σ_z apart. The second step adds ten LRG3–ELG1 pairs

(Section 7), for 107 pairs in all.

The $\Delta d_{\rm com} = 62$ Mpc data vector. With the added pairs the $\Delta d_{\rm com} = 62$ Mpc data vector holds 198 spectra: 15 cosmic-shear spectra, 76 autos and 107 cross spectra, in 53 bands each, 10,494 entries of which the mask keeps 5,674. Each added pair keeps the bands where both of its autos are kept, the rule every other pair follows. The Gaussian covariance rebuilt on this layout is positive definite and bitwise equal to the 97-pair covariance on the shared spectra. The exploration examples keep the 97-pair files (the `*_linear` dataset, mask, pair list and covariance), so their χ^2 values stay as committed.

The counts. Table 3 sorts every layout's pairs by kind. Most join a bin to the next one of its sample. The rest join an LRG and an ELG bin: the LRG3–ELG1 pairs of the shared volume (on $\Delta d_{\rm com} = 62$ Mpc, 22 from the first step and 10 from the second), one LRG2–ELG1 pair at $z = 0.8$ where LRG2 ends and ELG1 begins, and, on $\Delta z = 0.02$ and 0.05, one LRG3–ELG2 pair at $z = 1.1$. Each bin has 1 to 5 cross partners, 1 to 4 on $\Delta z = 0.2$; the first and last bins have one.

layout	bins	next bin	BGS–LRG1	LRG3–ELG1	LRG–ELG, other	added	total
$\Delta d_{\rm com} = 62$ Mpc	76	73	1	22	1	10	107
$\Delta z = 0.02$	90	87	1	43	2	0	133
$\Delta z = 0.05$	36	33	1	16	2	0	52
$\Delta z = 0.2$	10	7	1	4	1	0	13

Table 3: The measured cross spectra of each layout, by kind. "Next bin": two consecutive bins of one sample; BGS–LRG1: the pair across the BGS–LRG boundary at $z = 0.4$; "LRG–ELG, other": the LRG2–ELG1 pair at $z = 0.8$ and the LRG3–ELG2 pair at $z = 1.1$; "added": the LRG3–ELG1 pairs of the second step, whose raw ranges miss and whose blurred $n(z)$ overlap.

How to read Figure 2. Each block is one layout. Its upper panel draws every bin's raw $n(z)$ at unit peak, BGS and LRG in the upper lane, ELG in the lower one; two bins that touch share an edge. Its lower panel is the pair graph: a dot per bin at $\bar{z}_i$ and a segment per measured pair. The segments inside a row are the chain of neighbours; the segments between the rows are the LRG–ELG pairs, all inside the gray band where LRG3 and ELG1 share a volume, or at its edges. The dash-dotted vermillion segments of $\Delta d_{\rm com} = 62$ Mpc are the ten pairs of the second step. On $\Delta z = 0.02$ the LRG and ELG bins have the same edges, so each LRG3 bin touches three ELG1 bins: the one at its redshift and the two beside it. Figure 3 magnifies the corner where LRG2 ends and LRG3 and ELG1 begin.

7 What the rule means in redshift

Centre separations. A rule in terms of ranges can be compared with a rule in terms of centres: is there a separation $|\bar{z}_i - \bar{z}_j|$ below which every pair is measured and above which none is? Table 4 gives, per layout, the largest separation of a measured pair and the smallest of an unmeasured one.

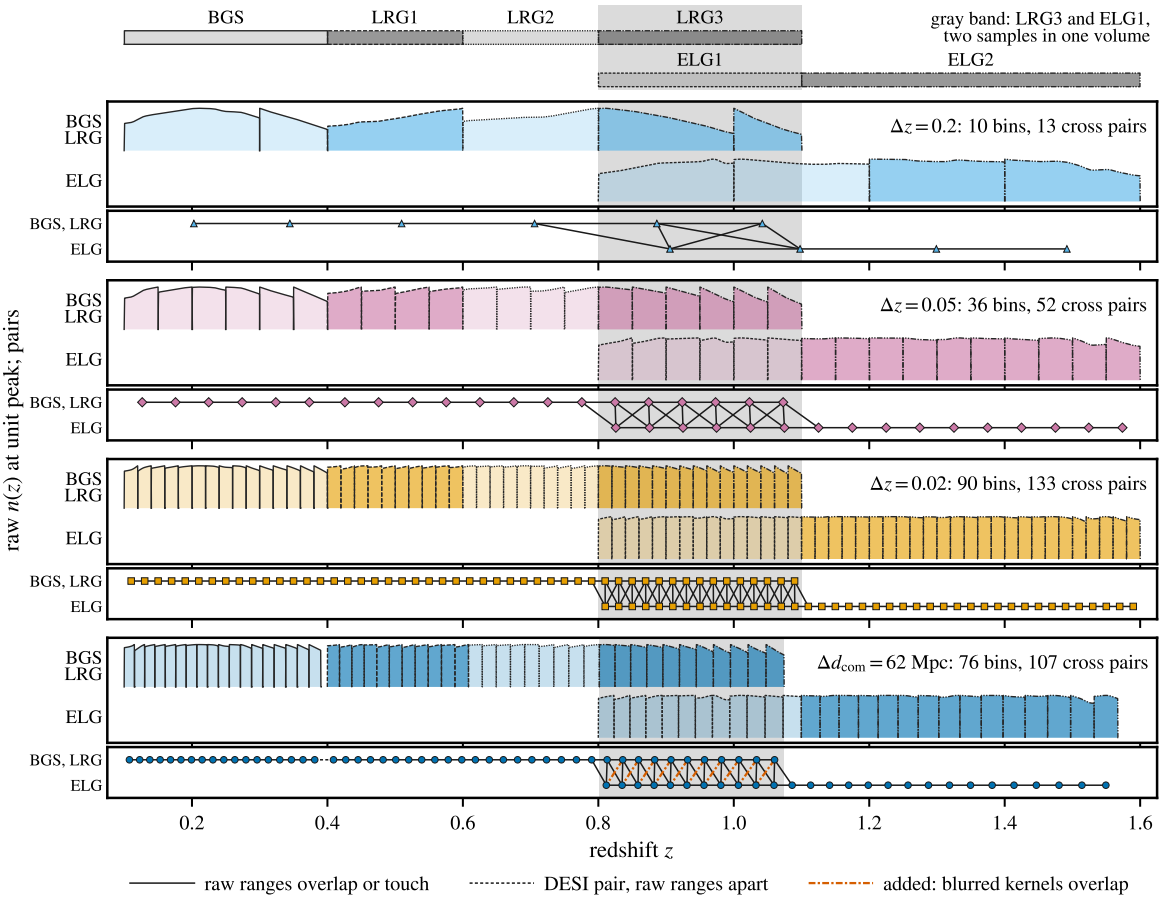


Figure 2: Every thin bin of the four layouts and the cross spectra the data vector measures. Upper panel of each block: each bin's raw $n(z)$ at unit peak, BGS and LRG in the upper lane, ELG in the lower; fill shade and outline style mark the six groups (header). Lower panel: a dot per bin at its mean redshift and a segment per measured pair; solid: raw ranges overlap or touch; dashed: the BGS–LRG1 pair of $\Delta d_{\rm com} = 62$ Mpc, whose ranges do not touch; dash-dotted vermillion: the ten LRG3–ELG1 pairs added because their blurred $n(z)$ overlap. Gray band: the redshifts where LRG3 and ELG1 both have bins.

Fixed redshift width: only adjacent bins. On the three fixed-Δz layouts the two columns are separated by about a factor of two: neighbours are one width apart, the next bin over is two widths apart. Any cut between the two columns, for example at 0.03 on $\Delta z = 0.02$, reproduces the rule. So on these layouts the rule means: only adjacent bins pair, where an LRG bin and an ELG bin with the same range count as adjacent (separation near zero). The edge bins of $\Delta z = 0.2$, 0.1 wide, shift the numbers but not the conclusion.

Fixed comoving width: no redshift cut works. On $\Delta d_{\rm com} = 62$ Mpc the columns cross: a measured pair is 0.035 apart and an unmeasured one 0.030. The reason is how a shell of fixed depth looks in redshift. The comoving distance is $\chi(z) = \int_0^z c\,dz'/H(z')$, with $H(z)$ the expansion rate, so a shell of depth $\Delta d_{\rm com}$ spans $\Delta z \approx \Delta d_{\rm com}\,H(z)/c$ in redshift. H grows with

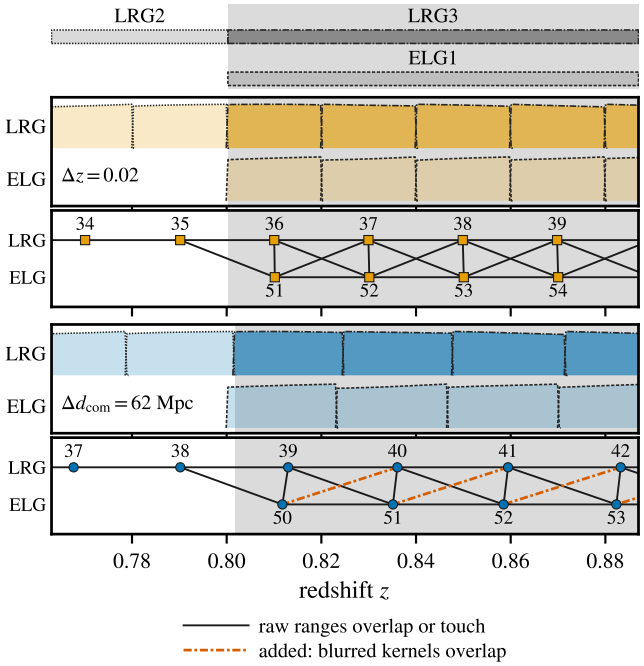


Figure 3: The corner of Figure 2 where LRG2 ends and LRG3 and ELG1 begin, on the two thin layouts, with every bin numbered in data-vector order. Solid: measured pairs whose raw ranges overlap or touch. Dash-dotted vermillion: the LRG3–ELG1 pairs of $\Delta d_{\rm com} = 62$ Mpc whose raw ranges miss by 3 to 5 grid steps and which the second step adds because their blurred $n(z)$ overlap (Section 5). A segment leaving the panel joins a bin outside it.

layout	bin width in z	largest measured	smallest unmeasured
$\Delta z = 0.2$	0.1–0.2	0.211 (LRG3 5, ELG1 8)	0.257 (LRG3 6, ELG2 9)
$\Delta z = 0.05$	0.05	0.053 (BGS 6, LRG1 7)	0.097 (LRG3 18, ELG1 22)
$\Delta z = 0.02$	0.02	0.020 (LRG3 48, ELG1 64)	0.040 (LRG3 45, ELG1 58)
$\Delta d_{\rm com} = 62$ Mpc	0.015–0.035	0.035 (ELG2 75, 76)	0.030 (BGS 1, 3)

Table 4: The separation $|\bar{z}_i - \bar{z}_j|$ of the mean redshifts of the farthest measured pair and of the closest unmeasured pair, with their groups and bin numbers. On $\Delta z = 0.2$ the bins at the ends of some samples are 0.1 wide.

z, so the same 62 Mpc covers 0.015 in redshift at $\bar{z} = 0.11$ and 0.035 at $\bar{z} = 1.55$.

Two examples make this concrete. BGS bins 1 and 3, one bin apart at $z \approx 0.12$, are only 0.030 apart in mean redshift and not paired; ELG2 bins 75 and 76, neighbours at $z \approx 1.53$, are 0.035 apart and paired. The rule is adjacency in comoving distance, and in redshift that adjacency stretches with z.

Misaligned LRG and ELG edges. The LRG3–ELG1 pairs of $\Delta d_{\rm com} = 62$ Mpc show a second effect. Each sample has its own bin edges: the LRG bins begin at $z = 0.400$ and the ELG bins at $z = 0.800$. In the shared volume the LRG edges sit 0.001–0.002 above the ELG edges (Figure 3, bottom). An LRG3 bin therefore overlaps the ELG1 bin at its redshift and

the one above it, and starts just above the end of the one below it. Bin 40 begins 0.0014 above the end of bin 50: 4 grid steps, more than the first step's 1.5, so the raw ranges miss, although the centres are only 0.024 apart. The blur closes that gap: the ten pairs of an LRG3 bin and the ELG1 bin ending just below it (bins 40 to 49 with 50 to 59) have raw ranges 3 to 5 grid steps apart, less than σ_z, and zero raw overlap, but blurred they overlap by 0.028–0.035, half as much as a neighbour pair (crosses in Figure 1; dash-dotted segments in Figures 2 and 3). At high ℓ they therefore correlate about half as strongly as neighbours do, and the second step of the rule adds them to the data vector. Each of the LRG3 bins 40 to 49 thus has three ELG1 partners, as on $\Delta z = 0.02$: two from the first step and one from the second, 32 LRG3–ELG1 pairs in all. With the ten added, no unmeasured pair of any layout overlaps by more than 10^{-6} after the blur; on the fixed-Δz layouts none did to begin with.

8 Which $n(z)$ each piece reads

The pair list reads the raw $n(z)$ in its first step, Eq. (5), and the blurred $n(z)$ at the default σ_v in its second. It is fixed with the data and does not move when σ_v is sampled. The model reads the blurred $n(z)$ in every spectrum it computes: the autos, the measured cross spectra and the stochastic templates. The covariance uses the blurred $n(z)$ at the default σ_v.

Together the two steps measure every pair whose kernels overlap in the model, and the DESI forecast's BGS–LRG1 boundary pair besides. On the fixed-Δz layouts the second step changes nothing: every pair whose blurred $n(z)$ overlap already touches in raw $n(z)$, and every measured pair overlaps by at least 0.003 once blurred.

9 What is not measured, and why that is a choice

Unmeasured spectra are not zero. Two thin bins one apart share no galaxies and, even blurred, no redshift range, so their cross spectrum vanishes in the Limber regime. At low ℓ it does not vanish: modes longer than a bin along the line of sight connect the two shells (Section 3), so their non-Limber cross spectrum is small but nonzero. The same holds for any pair beyond. After the second step of the rule no unmeasured pair of any layout has overlapping blurred kernels, so every unmeasured spectrum vanishes in the Limber regime and lives at low ℓ only.

Leaving a spectrum out is a choice of data, not an error. The likelihood compares each spectrum in the data vector with the model's prediction for that same spectrum, weighted by the covariance of the measured spectra. Leaving a pair out removes its rows from the data, the model and the covariance together. The remaining comparison is still a correct one; it uses less information. So the choice can cost precision, by the information the unmeasured spectra carry, but it cannot bias a fit: a bias would need a wrong prediction for a spectrum that is kept, and leaving another spectrum out changes no prediction.

10 How much do the low-overlap pairs carry?

The test. How much information do the weakly overlapping pairs contribute? The test keeps a cross spectrum only when its blurred overlap at the default dispersions reaches a floor,

$O_{ij} \geq O_{\min}$ (Eq. (4)), applied to *every* measured pair, the DESI-imported ones included; the auto spectra are always kept, a kept pair keeps its bands, and the covariance is rebuilt on the kept rows. The fit is model B + tilts on the pessimistic linear-bias prior, with α_2 free and the exploration σ_v widths (Section 2): the reference model of `notes/fog_parameter_degeneracy_study`. It is scored by the dark-energy figure of merit $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$, the inverse area of the marginalized ellipse of w_0 and w_a (the dark-energy equation of state $w(a) = w_0 + w_a(1 - a)$, a the scale factor), and by $\mathrm{FoM}(\Omega_m, 10^9 A_s)$, defined the same way for the matter density Ω_m and the primordial amplitude A_s. The fit carries the forecast's scale cut, computed on the paper model's fiducial (Section 2): bin i keeps wavenumbers below $k_{\max}/D(\bar{z}_i)$, with D the linear growth factor ($D(0) = 1$) and $k_{\max}$ always quoted at $z = 0$; Figure 4 shows $k_{\max} = 0.075$ and $0.2\,h/\mathrm{Mpc}$.

Which pairs drop. Every measured pair but one overlaps by at least 0.028 on $\Delta d_{\mathrm{com}} = 62$ Mpc, 0.043 on $\Delta z = 0.02$, 0.015 on $\Delta z = 0.05$ and 0.0034 on $\Delta z = 0.2$. The exception is the BGS–LRG1 boundary pair of $\Delta d_{\mathrm{com}} = 62$ Mpc, bins 18 and 19, whose blurred kernels overlap by 3.4×10^{-7} (Section 6). So the floors $O_{\min} = 10^{-6}$, 10^{-4} and 10^{-3} all drop that one pair and nothing else, and $O_{\min} = 10^{-2}$ additionally drops 10 of the 13 pairs of $\Delta z = 0.2$, whose overlaps all sit below 10^{-2} (Section 5.5).

What it costs. Dropping the boundary pair changes $\mathrm{FoM}(w_0, w_a)$ by a factor 0.999 at $k_{\max} = 0.2\,h/\mathrm{Mpc}$ and nothing at the smaller cuts, and every other layout is unchanged at every floor up to 10^{-3} (Figure 4). At 10^{-2} the $\Delta z = 0.2$ figures of merit fall to 0.82 of the full vector's for (w_0, w_a) and to 0.65–0.72 for $(\Omega_m, 10^9 A_s)$.

The reading. The only listed pair with overlap below 10^{-3} is the boundary pair, and it costs 0.1% of one figure of merit at the largest cut. The list holds no pair between 3.4×10^{-7} and 3.4×10^{-3} – on the layouts other than $\Delta z = 0.2$ the next-smallest overlap is 0.015–0.043 – so this test cannot say where between those overlaps a pair begins to matter. And a small overlap does not mean little information: the 13 pairs of $\Delta z = 0.2$ overlap by only 0.3–0.9%, yet removing ten of them costs up to 35% of $\mathrm{FoM}(\Omega_m, 10^9 A_s)$.

11 Open issues

- **The BGS–LRG1 boundary pair of $\Delta d_{\mathrm{com}} = 62$ Mpc.** It is measured, as the DESI forecast measures it, although its ranges are 0.010 apart and its blurred kernels barely overlap (3×10^{-7}, below the second step's threshold); its signal is the low-ℓ non-Limber correlation alone.

- **The list does not follow σ_v.** It is built at the default dispersions. A sampled σ_v changes every kernel overlap but not which pairs are measured. Within two prior widths of the defaults no unmeasured pair of any layout reaches an overlap of 10^{-4}.

- **The pair list at the σ_v prior edge.** Just beyond those two prior widths the thin layouts' bins two apart begin to overlap by more than 10^{-6} (`notes/fog_parameter_degeneracy_study`, on the pair rule at the edges of the σ_v prior). Building the list at

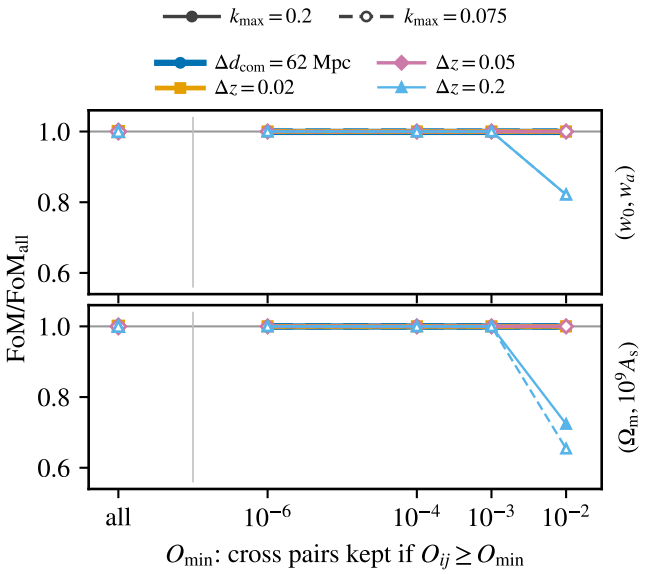


Figure 4: The figures of merit of the fit of Section 10 with only the cross pairs whose blurred overlap at the default dispersions reaches $O_{\min}$, over the full data vector's, against $O_{\min}$; "all", every measured pair, sits left of the separator. Upper panel: FoM(w_0, w_a); lower: FoM$(\Omega_m, 10^9 A_s)$; the panels share the y axis. Color, line width and marker: the layout; solid, filled markers: $k_{\max} = 0.2\,h/\mathrm{Mpc}$; dashed, open: 0.075; gray: one. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

the prior edge, which adds those pairs and needs new spectra, covariances and derivative caches, is open.

Reproducing the figures. `plots/make_cross_correlations_data.py` reads the project's `data/` and `likelihood/` files and writes the `data.json` of the first three figure folders, with every number quoted for them here; `plots/pair_threshold/data.json` is written by the Fisher notebook `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb` (section 11, item 7). Each folder's `plot_<name>.py` redraws its figure from that file alone.

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